



Tobias Rosenbaum

Associate Director E2E SCM Systems

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Profile

IT and quality leader with 15+ years in regulated GxP environments (GCLP/GMP). Long experience in cross domain work, bridging business, technology, and compliance at BioNTech, e.g. by implementing worlds fastest QC LIMS for Comirnaty, as well as working with IVAC leadership to streamline IVAC system landscape. Proven record delivering risk-based quality strategies, audit/inspection readiness, vendor oversight, and governance across complex validated systems at BNT. Clinical-adjacent leadership includes GCLP team management, establishing the clinical front end for individualized drug production interfacing with IRT/EDC, and cross-functional RBQM-style risk governance.

Professional Experience

Dec 2024 – present
Mainz, Germany

ASSOCIATE DIRECTOR E2E SUPPLY CHAIN SYSTEMS, BIONTECH SE, *BIONTECH SE*

Lead cross-functional teams establishing cross domain data flows for clinical orders, chain of Identity (COI) and chain of Custody (COC) compliance.

- Establish critical cross domain data flows between gclp and gcp areas, integrating data from clinic facing systems like IRT and EDC systems into the iNest process.
- IT business partner for CTSM, working closely with IMP-Management and patient operations.
- Implemented streamlined validation approach for SAAS based on GAMP5 and CFR Part11 in cooperation with Global System-QA.
- Lead team to spearhead implementing CTSM/SCM solutions for BNT327, e.g. E2E Inventory Management in collaboration with Chinese sites (Nantong)
- Manage AI automation project using Databricks and large language models to streamline validationdocument generation, aiming for over 80% efficiency improvement.

May 2021 – Nov 2024
Mainz, Germany

ASSOCIATE DIRECTOR DIGITAL SOLUTIONS QUALITY SYSTEMS ARCHITECTURE, *BIONTECH SE*

Led team of architects developing enterprise-scale IT solutions for quality control, quality assurance and clinical labs.

- Defined cross-domain process and data flow models enabling traceability and ALCOA+ principles end to end in cooperation with Head of GCP-QA, DD&I and the data value office.
- Implemented ISO 25010-based quality framework for vendor selection and system design, used in several software implementation projects.
- Took over leading role in the company's FDA audit preparation program's system workstream, leading to zero critical system related findings in the following audits.

Oct 2018 – May 2021
Mainz, Germany

SYSTEM ENGINEER, BIONTECH SE

Established processes for GxP compliant applications and led major implementation projects.

- Led record-breaking global QC LIMS implementation (7 months, 3 sites) as solution architect, directly enabling quality control processes supporting €19 billion revenue impact, incl. validation strategy, DI controls (e.g., audit trail configuration, access management, backup and recovery)
- Led cross-functional initiatives with CSV-QA in establishing GxP-compliant application processes, significantly reducing the time needed for the approval of documents.
- Led GCLP application team through critical transition period during COVID vaccine development
- Standardized change control and release validation across LIMS, increasing consistency of reviews and DI documentation.

Aug 2012 – Sep 2018
Wiesentheid, Germany

PROFESSIONAL SYSTEM ENGINEER / IT QUALITY OFFICER, *Bayerischer Blutspendedienst gGmbH*

and Managed full lifecycle of 30+ applications processing 7,500 blood products daily, pioneering data-driven insights.

- Served as IT Quality Officer, representing the company in both internal and external system related audits.
- Wrote several SOPs with regards to computerised system validation, data integrity and systems engineering.
- Managed full lifecycle of software projects for over 30 key applications involved in production and QP-release of over 7,500 blood products daily, including investigating deviations and owning system related CAPAs.
- Executed QC-LIMS implementation with detailed V-model validation, setting a new vendor benchmark.

Jun 2011 – Jul 2012
Penzberg, Germany

SYSTEM ANALYST, ROCHE DIAGNOSTICS GMBH

- Created and owned QA dashboard, aggregating data from 4 MES systems for batch release, ensuring GMP compliance and data integrity across manufacturing data flows.
- Validated MES for Herceptin production across 4 facilities (test design/execution, documentation review).
- Responsible for managing system related Deviations and CAPAs

Education

Oct 2005 – Feb 2011
Würzburg, Germany

German Diploma (Biology/Life Sciences), *Julius-Maximilians-Universität*

Focus: Computational Biology, Neurobiology, Bioinformatics Thesis: Pattern recognition in brain waves of honey bees

Sep 2002 – Jan 2005
Frankfurt a.M., Germany

Computer System Developer, Proadis Partner für Bildung GmbH Completed 3-year program in 2.5 years due to exceptional performance

1992 – 2001
Mainz, Germany

Allgemeine Hochschulreife (Abitur), Theresianum Gymnasium Mainz

Projects

2022 – 2023

Clinical Lab Blueprint, Domain Architect

- Redesigned software landscape for individualized mRNA vaccine production in close collaboration with IVAC Leadership and business stakeholders
- Reduced target system landscape from 25 legacy systems to 4 central domain-specific systems, enabling faster adaption for new sites like Upstream Innovation Center
- Defined integration standards improving data integrity and reducing turnaround time together with Head of GCP QA, DD&I and data value office

2020 – 2021

BioNTech Global QC LIMS Implementation, Solution Architect

- Coordinated validation strategy across manufacturer, QA, validation manager, and IT teams, reducing validation efforts bei roughly 80%.
- Delivered project in 7 months from start to multi-site rollout.
- Enabled quality control for increasing Comirnaty production, supporting €19bn revenue.
- Established company standards for gxp user requirements and validation documentation with Enterprise IT.

2014 – 2017

Major release update of blood banking solution Edgeblood, Project Lead

- Planned and conducted validation for major system update impacting all critical production processes.
- Created risk based test plan, test strategies and test report in cooperation with QA and qualified person.
- Represented implementation and validation of update in both customer- and regulatory audits.

Core Competencies

GxP Compliance & Computer System Validation (13+ years)

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Team Leadership & Cross Functional Project Management

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GxP System Ownership & Governance

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Stakeholder Management & C-Level Consulting

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BNT Lightspeed experience

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Delivering great results in the face of challenging timelines

Digital Strategy & System Architecture

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Certificates

- Expert for Computersystemvalidation
- ITIL Foundation Level

- Certified Professional Software Architect Foundation Level

Languages

- German
- English

Interests

- Photography
- Woodworking and DIY
- Agentic Systems